

torch.signal.windows.hamming#

<https://pytorch.org/docs/stable/generated/torch.signal.windows.hamming.html>

Description:

Computes the Hamming window. The Hamming window is defined as follows: The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True. M (int) – the length of the window. In other words, the number of points of the returned window. sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

Function Signature

```
torch.signal.windows.hamming(M, *, sym=True, dtype=None,
layout=torch.strided, device=None, requires_grad=False)
[source]#
```

Parameters

Keyword Arguments

Return type

Parameters

Keyword

`sym` (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True. `alpha` (float, optional) – The coefficient α in the equation above. `beta` (float, optional) – The coefficient β in the equation above. `dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`). `layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`. `device` (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and t

Returns:

Tensor

Code Examples

```
>>> # Generates a symmetric Hamming window.
>>> torch.signal.windows.hamming(10)
tensor([0.0800, 0.1876, 0.4601, 0.7700, 0.9723, 0.9723, 0.7700,
        0.4601, 0.1876, 0.0800])

>>> # Generates a periodic Hamming window.
>>> torch.signal.windows.hamming(10, sym=False)
tensor([0.0800, 0.1679, 0.3979, 0.6821, 0.9121, 1.0000, 0.9121,
        0.6821, 0.3979, 0.1679])
```

Detailed Documentation

Documentation

Computes the Hamming window.

The Hamming window is defined as follows:

The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True.

M (int) – the length of the window. In other words, the number of points of the returned window.

sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

`alpha` (float, optional) – The coefficient α in the equation above.

`beta` (float, optional) – The coefficient β in the equation above.

`dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`).

`layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`.

`device` (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

`requires_grad` (bool, optional) – If autograd should record operations on the returned tensor. Default: False.